TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Cell-Free Biomanufacturing Platform Using DNA Nanostructure Scaffolds			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/Medical		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Microbial and Enzyme Catalysts	Polymeric Materials	Nanomaterial Technologies
Concept of Proposed Technology	We propose the development of a cell-free biomanufacturing platform that uses programmable DNA nanostructures as molecular scaffolds to spatially organize enzyme cascades with nanometer precision. AI-generated aptamers serve as molecular "Velcro" to attach specific enzymes at defined positions on DNA scaffolds, creating optimized multi-enzyme assembly lines that enhance reaction efficiency through substrate channeling and controlled stoichiometry. Unlike traditional enzyme immobilization, which randomly attaches enzymes to surfaces, our platform enables rational design of enzyme spatial arrangements, allowing precise control over multi-step biosynthetic pathways without living cells. Our team has extensive experience with tile-based DNA nanostructure design and assembly, having developed star-shaped DNA scaffolds displaying aptamers in precise 2D patterns matching viral epitope spatial arrangements [1,2]. In this proposed work we will leverage AptaBLE [3], a generative AI model that we previously developed to produce the aptamers that will anchor the enzymes at defined positions on the DNA scaffolds. AptaBLE generated aptamers have been demonstrated to have nM Kds across a diverse (CD117, CD28, CD4, TIGIT) set of protein targets [3]. This platform would overcome traditional SELEX limitations that require months of iterative selection to identify			

	ideal aptamers. We will also leverage our experience with engineering flexible linkers to modulate the activity of immobilized enzymes. Previous work in our group demonstrated that optimized linker length and flexibility dramatically enhanced catalytic activity of immobilized enzymes [4]. These findings translate well to DNA scaffolds, where we incorporate flexible single-stranded DNA regions at strategic positions for local flexibility while maintaining overall rigidity. Positioning enzymes at optimal distances on DNA platforms can result in 5-10x improvements versus freely diffusing enzymes [5,6]. These systems enable cofactor recycling (e.g., NAD+/NADH regeneration), dramatically reducing expensive cofactor requirements [7]. Cell-free platforms provide key advantages: (1) no competing metabolic pathways, (2) handling cell-toxic substrates/intermediates, (3) direct control over pH, temperature, cofactor concentrations, (4) rapid prototyping without genetic engineering timelines, (5) simplified purification without cellular debris.
Rationale for Joint R&D Project	Korean pharmaceutical and fine chemical SMEs face increasing pressure to adopt sustainable, cost-effective manufacturing processes. Traditional whole-cell fermentation systems are limited by slow production cycles, competing metabolic pathways, and difficulty handling toxic intermediates. Current cell-free enzyme systems offer faster reactions but suffer from random enzyme positioning, lack of spatial organization for substrate channeling, and poor biocatalyst reusability. This project combines RPI's expertise in DNA nanostructure engineering, AI-driven aptamer design, enzyme immobilization, and bioanalytical characterization with Korean partners' pharmaceutical manufacturing capabilities, Asian market regulatory knowledge, and high-value target molecule identification. This project combines RPI's expertise in DNA nanostructure engineering, AI-driven aptamer design, enzyme immobilization, and bioanalytical characterization with Korean partners' pharmaceutical manufacturing capabilities, Asian market regulatory knowledge, and high-value target molecule identification. RPI brings proven core technologies: (1) Demonstrated expertise in designer DNA nanostructure synthesis for biomolecular organization, including star-shaped architectures with nanometer-scale positioning precision validated for viral inhibition; (2) Computational protein engineering using PROSS achieving 3-fold enzyme activity improvements for complex sulfotransferases in microbial biosynthetic pathways [9]. This enabled complete biosynthesis of sulfated chondroitin in E. coli with 96% product sulfation [9]; (3) Validated strategies for enzyme surface immobilization using peptide linker engineering [12]. Our bacteriolytic enzyme work demonstrated flexible peptide linkers maintain high catalytic activity across multiple reuse cycles (>99.5% bacterial killing, five reuse cycles) under harsh conditions including industrial surfactants [12]; (4) AI-powered aptamer generation (AptaBLE platform) producing experimentally validated aptamers with Kd from 31 nM to low micromolar, plus retrospective SELEX analysis identifying previously overlooked high-affinity binders (sub-50 nM) [10]; (5) Advanced RNA design (EvoFlow-RNA) enabling unconditional generation of diverse RNA sequences with native-like properties and conditional generation through motif scaffolding with RMSD <1 Å [11].

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	Korean SME partners contribute pharmaceutical manufacturing infrastructure, regulatory expertise for biologics, quality control systems meeting international standards, and established distribution channels in Asian pharmaceutical markets. Their experience scaling laboratory processes to commercial production ensures technology development remains focused on commercially viable applications. Key synergies: • RPI designs/validates DNA scaffolds; Korean partners identify high-value pharmaceutical targets • RPI develops AI-generated aptamers; Korean partners optimize fermentation-free workflows • RPI characterizes cascade efficiency at lab scale; Korean partners evaluate manufacturing economics and scale-up • Joint IP development with complementary markets (RPI: Americas/Europe; Korean: Asia-Pacific) This accelerates commercialization by combining cutting-edge research with industrial manufacturing expertise.
Key Words	1. Biomanufacturing 2. Cell free manufacturing 3. Nanotechnology 4. DNA origami scaffolds 5. Enzyme cascade optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

Primary Objective:
Develop and validate a DNA scaffold-based cell-free biomanufacturing platform achieving >5-fold cascade efficiency improvements compared to free enzyme systems for pharmaceutical intermediate synthesis.

Specific Technical Objectives:

1. Design and validate 3-5 DNA scaffold architectures (linear, 2D planar, 3D frameworks) with defined geometries optimized for multi-enzyme organization using tile-based assembly

2. Generate high-affinity aptamers (target Kd <100 nM) for 10+ target enzymes using AptaBLE platform with experimental validation via SPR

3. Develop and optimize flexible linkers while maintaining >80% enzyme activity when attached to DNA scaffolds versus free enzyme

4. Establish quantitative distance-activity relationships for 2-4 enzyme cascade systems, identifying optimal interenzyme spacing (10-25 nm) maximizing substrate channeling efficiency

5. Implement cofactor recycling systems (NAD+/NADH, ATP/ADP) reducing cofactor consumption >10-fold compared to stoichiometric addition

6. Demonstrate cell-free synthesis of 2-3 Korean market pharmaceutical intermediates with >5× productivity improvements over incumbent technologies

7. Develop reusable scaffold systems maintaining >80% activity through >10 use cycles with economically viable recovery/regeneration protocols

8. Scale validated processes to 1 L pilot production with documented productivity (g/L/day), economics ($/kg product), and quality (>95% purity) meeting pharmaceutical standards

R&D Target Specifications

The following quantitative performance targets will be achieved and validated through the specified measurement methods:

Performance Metric	Target Specification	Measurement Method
Aptamer binding affinity	Kd<50nM for target enzymes	Surface plasmon resonance
DNA scaffold positioning accuracy	Sub-5nm spatial resolution	Atomic force microscopy
Cascade efficiency improvement	10-100× vs. free enzyme controls	Kinetic analysis, product quantification
Enzyme reusability	>80% activity after 10 cycles	Activity assays, enzyme quantification

	Substrate channeling efficiency	>80% intermediate capture	Metabolite flux analysis
	Pathway engineering timeline	<1 month per new cascade	Process documentation
	Product yield	>2-fold improvement vs. current methods	Analytical chemistry (HPLC, LC-MS)
	Cost reduction	>50% reduction in cost-per-kg product	Techno-economic analysis
	Scale demonstration	100 mL-1 L reaction volumes	Pilot-scale validation
	Temperature stability	Maintain activity 4-25°C for 72h	Stability testing

R&D Plan

The 24-month project is structured in three phases, progressing from laboratory validation (TRL 4) through pilot-scale demonstration (TRL 6) to customer validation and commercialization readiness (TRL 7):

Phase 1: Core Technology Development & Integration (Months 1-6, TRL 4→5)

Key Tasks:

- Task 1.1: Use the AptaBLE AI-driven aptamer generation pipeline to generate and screen aptamers for 3-5 initial target enzymes
- Task 1.2: Design and synthesize DNA nanostructure scaffolds - Create library of 5+ scaffold architectures with different geometries, validate self-assembly and structural integrity, characterize positioning accuracy via AFM/cryo-EM
- Task 1.3: Integrate aptamers with DNA scaffolds - optimize aptamer placement strategies, verify enzyme binding capacity and orientation control
- Task 1.4: Initial cascade assembly and testing - Construct 2, and 3-step enzyme cascades for model systems, measure efficiency improvements vs. free enzymes, optimize reaction conditions and enzyme ratios

Milestones:

- M1: AI aptamer pipeline operational with >80% prediction accuracy (Month 1)
- M2: DNA scaffold library validated with sub-5nm positioning (Month 3)
- M3: First functional enzyme-scaffold assembly with confirmed activity (Month 4)
- M4: 10× efficiency improvement demonstrated in model cascade (Month 6)

Phase 2: Pathway Engineering & Optimization (Months 6-12, TRL 5→6)

Key Tasks:

- Task 2.1: Partner-directed pathway selection - Work with Korean SME to identify 2-3 commercially relevant target molecules, assess market requirements and regulatory considerations, define success criteria for each pathway
- Task 2.2: Cascade design and construction - Design multi-enzyme cascades (3-5 enzymes) for target pathways, generate required aptamers, assemble and optimize scaffold-enzyme configurations
- Task 2.3: Performance optimization - Conduct systematic studies of enzyme loading, spacing, and stoichiometry, optimize substrate/cofactor concentrations, improve reaction kinetics through iterative design

- Task 2.4: Recovery and reuse protocols - Develop methods for enzyme-scaffold separation and recovery, test stability over multiple reaction cycles, implement regeneration procedures as needed
- Task 2.5: Scale-up to 100mL-1L volumes - Adapt laboratory protocols to pilot scale, address mixing and mass transfer considerations, validate performance consistency at larger scale

Milestones:

- M5: First commercial pathway operational with >20× improvement (Month 8)
- M6: Enzyme recovery achieving >90% activity retention (Month 8)
- M7: 5L pilot-scale demonstration successful (Month 10)
- M8: Second commercial pathway validated (Month 12)

Phase 3: Validation & Commercialization Preparation (Months 12-24, TRL 6→7)

Key Tasks:

- Task 3.1: Customer validation studies - Provide samples to SME partner for internal testing, conduct side-by-side comparisons with current methods, gather feedback on performance and usability
- Task 3.2: Manufacturing process development - Establish protocols for scaffold/aptamer production at scale, optimize enzyme sourcing and quality control, develop standard operating procedures
- Task 3.3: Techno-economic analysis - Calculate cost-per-kg for target products, identify cost drivers and optimization opportunities, project manufacturing economics at commercial scale
- Task 3.4: Regulatory and IP strategy - Assess regulatory pathways for target applications, file patent applications on key innovations, develop commercialization roadmap
- Task 3.5: Platform generalization - Demonstrate rapid pathway engineering capability (<1 month), test additional enzyme classes and reaction types, create design guidelines and documentation

Milestones:

- M9: Positive customer validation results (Month 15)
- M10: Manufacturing cost <50% of current methods (Month 18)
- M11: Third pathway demonstrated with <1 month development time (Month 20)

M12: Commercialization-ready platform with complete documentation (Month 24)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead development of AI-powered aptamer generation technology, including machine learning model training, validation, and continuous improvement based on experimental feedback • Design and engineer DNA nanostructure scaffolds with optimized geometries for different enzyme cascade configurations, leveraging existing expertise in DNA origami and biosensor development • Conduct fundamental research on enzyme spatial organization, substrate channeling mechanisms, and structure-function relationships to guide rational platform design • Provide access to advanced characterization facilities (AFM, cryo-EM, spectroscopy) and computational resources for modeling and simulation • Train Korean SME personnel in platform operation, troubleshooting, and customization through on-site visits, virtual training sessions, and detailed documentation • Lead intellectual property development including patent filing, technology disclosures, and protection of proprietary methods • Publish scientific findings in peer-reviewed journals to establish technology credibility and attract additional commercial interest • Contribute 1-2 technicians as full-time project personnel, plus PI time commitment of 20% effort
Expected R&R of Korean SME Partner	• Identify commercially valuable target molecules and biosynthetic pathways based on market analysis, customer needs, and regulatory feasibility for Korean pharmaceutical/chemical markets • Provide commercially relevant enzymes, substrates, and analytical standards for pathway development and testing, leveraging existing supplier relationships • Conduct pilot-scale validation studies in manufacturing-relevant environments, including GMP considerations and process consistency testing • Lead customer engagement activities including sample distribution, application testing, feedback collection, and technical support • Develop manufacturing protocols for platform components (scaffolds, aptamers) suitable for commercial-scale production, identifying suppliers and establishing quality control procedures • Perform techno-economic analyses comparing platform economics to existing manufacturing methods, identifying cost reduction opportunities and market positioning strategies • Navigate Korean regulatory requirements for biomanufactured products, including preparation of technical dossiers and engagement with regulatory authorities as appropriate • Invest in necessary equipment and facilities for pilot-scale operation, including bioreactors, purification systems, and analytical instruments • Contribute 2-3 full-time technical staff including process development engineers, analytical chemists, and project management personnel

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** • Programmable biomanufacturing platform enabling rational design of multi-enzyme cascades without living cells • Validated AI-aptamer generation system reducing enzyme binding molecule development time from months to days • Demonstration of 10-100× efficiency improvements in pharmaceutical intermediate synthesis through spatial enzyme organization • Proven enzyme recovery and reuse protocols achieving >80% activity retention, dramatically reducing biocatalyst costs • Modular platform architecture enabling rapid pathway engineering (<1 month) for diverse target molecules • Comprehensive technology package including design tools, protocols, manufacturing specifications, and operational guidelines ready for commercial deployment **Commercialization Potential** • Korean SME partner positioned to offer differentiated biocatalysis services that major CDMOs cannot easily replicate, capturing premium pricing in specialty chemical markets • Estimated cost reduction of >50% for target pharmaceutical intermediates, enabling competitive advantage in global markets • Platform applicable to multiple high-value product categories: chiral building blocks, natural product derivatives, specialty polymers, diagnostic reagents • Projected market opportunity of $10-50M annually within Korean pharmaceutical/chemical sector within 3-5 years post-project • Foundation for spinoff applications in biosensors, diagnostics, environmental remediation using the same DNA nanostructure platform technology • Establishment of Korea-US collaboration framework serving as model for future joint technology development initiatives **Broader Industrial and Economic Impact** • Advance sustainable manufacturing through replacement of harsh chemical processes with enzymatic alternatives, reducing solvent use and waste generation • Strengthen Korea's position in biomanufacturing technology, creating competitive advantage in the transition toward bio-based economy • Generate high-quality technical employment opportunities in advanced biotechnology for Korean scientists and engineers • Enable production of pharmaceutical intermediates and fine chemicals currently not economically feasible with conventional methods • Establish RPI-Korea SME partnership as long-term collaboration platform for continued technology development and knowledge exchange • Create publication and patent portfolio enhancing research visibility and attracting follow-on industry partnerships and funding opportunities • Provide training opportunities for graduate students and postdocs in translational research bridging academic innovation and industrial application

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization